CozyStudy – Test Case Document

Project: CozyStudy – Aesthetic Gamified Study Sanctuary (Python 3 + Tkinter) **Prepared by:** [Your name] | **Roll no.:** [Your roll no.] | **Course:** [Course name] **Submitted to:** [Professor's name] | **Date:** 23 October 2026 | **Version:** 2.0 (after review feedback)

1. Purpose and scope

This document records how CozyStudy was tested after the project review. The reviewer raised two points:

1. On some screens the button background is very light and the text is also light, so the text is hard to read.
2. The application should be tested properly, including a way to reach the final state of the Cozy Room (every charm unlocked, with data on every screen).

This document covers the defects found, the fixes, the automated tests, the colour-contrast audit, the manual test cases, and a dedicated end-state test for the fully decorated room.

2. Test environment

Item	Value
Operating system	macOS 15
Python version	3.14
Libraries	Python standard library only (Tkinter, unittest, json, dataclasses)
Themes tested	Sakura Blossom, Matcha Study, Lavender Haze, Midnight Lo-Fi
Data files	cozystudy_data.json (normal mode), cozystudy_data_dev.json (dev mode)

How to run the tests

- All automated tests: `python3 -m unittest -v`
- Colour-contrast audit: `python3 contrast_audit.py`
- Application: `python3 main.py`
- Application in developer (test) mode: `python3 main.py --dev`

- GUI tests on a machine without a screen: `xvfb-run python3 -m unittest test_room_gui -v`

3. Test approach

Level	What it checks	Where
Unit / backend tests	Pure logic: rewards, integrity formula, charm rules, task bonus, streaks, analytics, save and load	test_app.py, test_end_state.py
Static contrast audit	Every foreground/background colour pair in the real source code, in all 4 themes (WCAG 2.1)	contrast_audit.py, test_contrast.py
Automated GUI tests	Full room draws all 12 charms in every theme; every visible label and button is readable	test_room_gui.py
Manual test cases	Screens, themes, anti-cheat timer, tasks, wardrobe, persistence, dev mode	Section 8

The application follows a functional-core design: the business logic is pure functions on immutable data, so it can be tested without opening a window. This is what makes the backend end-state test possible.

4. Review feedback and defects fixed

ID	Issue	Cause	Fix	Verified by
DEF-01	Application crashed on launch	TaskView never stored the state or its four callbacks	Stored them in the constructor	TC-M-01, test_room_gui
DEF-02	Button text hard to read (light on light)	On macOS, native tk.Button ignores the background colour, so light theme text sat on a light button	New FlatButton widget (widgets.py) that honours colours on every OS; all 24 buttons converted	TC-C-03, TC-M-02 to TC-M-05

ID	Issue	Cause	Fix	Verified by
DEF-03	Navbar title and charm-count pill low contrast (1.7 to 2.7 : 1)	They used the accent_hover colour on a light card	Now use brand_fg and coin_pill_fg tokens	TC-C-01
DEF-04	Changing theme restyled only some screens (mixed palettes)	Shell and Settings were built once and never restyled	Shell is restyled and Settings is rebuilt on theme change	TC-M-06
DEF-05	Anti-Cheat mode in Settings did not reach the timer	Timer's drop-down was never connected to saved settings	Timer reports changes and syncs from settings	TC-M-17, TC-M-18
DEF-06	No way to reach the full-room end state for testing	No test mode	Developer mode (--dev) with Unlock Everything, separate save file, and backend reducers	TC-B-01 to TC-B-10, TC-M-08 to TC-M-14
DEF-07	Demo sessions for Today could fall on yesterday before 5 a.m.	Sessions were placed at now minus N hours	Sessions are placed at fixed hours of the calendar day	TC-B-06

Regression tests also cover two defects fixed earlier: the streak date resetting on every load (TC-B-26) and the task bonus being awarded twice (TC-B-18).

5. Colour-contrast audit

The audit reads the real source files, finds every widget that sets both a text colour and a background colour, and checks each pair in all four themes. The thresholds are 4.5 : 1 for normal text and 3 : 1 for large text.

	Colour pairs found	Checks (pairs x 4 themes)	Failures	Raw tk.Button left
Before fixes (original files)	126	504	6	9
After fixes	131	524	0	0

The six original failures were the navbar title and the charm pill in the Sakura, Matcha and Lavender themes. The nine raw buttons are the ones affected by the macOS background-colour limitation (DEF-02).

6. Automated test cases

Result of the headless run on Python 3.12: **34 tests, 34 passed.** The 3 GUI tests are skipped without a display and are run on the student's machine (see Section 7).

6.1 Backend end-state tests (test_end_state.py)

ID	Test	Verifies	Expected
TC-B-01	all_12_charms_unlocked_and_equipped	After Unlock Everything all 12 charm ids exist, unlocked and equipped	Pass
TC-B-02	every_category_present	Desk, wall, pet and floor charms are all present	Pass
TC-B-03	original_state_untouched	The starting state is not changed (immutability)	Pass
TC-B-04	frozen_models	Assigning to a frozen dataclass field raises an error	Pass
TC-B-05	rich_data_for_every_screen	9999 charms, streak of 15 or more, all 5 subjects, 14 or more days of sessions, completed, overdue, due-today and future tasks	Pass
TC-B-06	today_timeline_never_empty	Today's timeline always has 3 sessions	Pass

ID	Test	Verifies	Expected
TC-B-07	analytics_consistent_with_sessions	Totals, switches, integrity and 7-day trend match the sessions	Pass
TC-B-08	json_roundtrip_equals_original	Full-room state survives state to JSON to state unchanged	Pass
TC-B-09	dev_reset_keeps_settings	Reset to Demo restores starter data but keeps the chosen theme and mode	Pass
TC-B-10	dev_helpers	Add 1000 charms and Simulate Session update coins and sessions	Pass

6.2 Reward, charm, task, streak and storage rules (test_end_state.py)

ID	Test	Verifies	Expected
TC-B-11	reward_table	Charm reward for a 1500-second session at integrity 100, 95, 90, 89, 80, 79, 60, 59, 0 in honest and strict mode (20/22, 20/22, 20/22, 15/18, 15/18, 8/10, 8/10, 3/6, 3/6)	Pass
TC-B-12	integrity_edges	0 total seconds gives 100 percent; active above total is capped at 100; 0 active gives 0; 1125 of 1500 gives 75	Pass
TC-B-13	unlock_needs_enough_coins	Unlock fails and state is unchanged when coins are short	Pass
TC-B-14	unlock_deducts_exact_cost	Unlock succeeds, deducts the exact cost, charm becomes unlocked and equipped	Pass
TC-B-15	cannot_unlock_twice	A second unlock does not charge again	Pass
TC-B-16	toggle_equip_ignored_for_locked	Equip does nothing on a locked charm	Pass

ID	Test	Verifies	Expected
TC-B-17	toggle_equip_flips_unlocked	Equip toggles an unlocked charm	Pass
TC-B-18	no_double_bonus	Completing a finished task again gives 0 bonus	Pass
TC-B-19	bonus_range	Task bonus is 20 (late), 35 (on time) or 45 (on time and high priority)	Pass
TC-B-20	consecutive_days_increment	Streak goes up by 1 on the next day	Pass
TC-B-21	same_day_unchanged	A second session the same day keeps the streak	Pass
TC-B-22	gap_resets_to_one_and_keeps_best	A missed day resets the streak to 1 and keeps the best streak	Pass
TC-B-23	full_room_survives_save_and_load	Saving and loading the full-room file gives an identical state	Pass
TC-B-24	dev_mode_uses_separate_file	Dev mode saves to cozystudy_data_dev.json, normal mode to cozystudy_data.json	Pass
TC-B-26	streak_survives_save_and_load	Streak and last-active date are not reset on load	Pass

6.3 Original suite (test_app.py)

ID	Test	Verifies	Expected
TC-B-27	immutability	Task and charm cannot be changed in place	Pass
TC-B-28	functional_combinators	pipe, curry, map_tuple, filter_tuple	Pass
TC-B-29	pure_reducers	complete_task and add_task leave the original state untouched	Pass
TC-B-30	focus_integrity_calculation	Anti-cheat integrity formula	Pass

ID	Test	Verifies	Expected
TC-B-31	analytics_pipelines	Aggregate, timeline and 7-day trend outputs	Pass
TC-B-32	storage_serialization	State survives a dict round trip	Pass

6.4 Colour-contrast tests (test_contrast.py)

ID	Test	Verifies	Expected
TC-C-01	all_widget_pairs_in_source_pass	All 131 colour pairs in the source meet WCAG in all 4 themes (524 checks)	Pass
TC-C-02	key_theme_tokens	14 key text/background token pairs meet 4.5 : 1 in every theme	Pass
TC-C-03	no_raw_tk_button_left	No raw tk.Button remains; all buttons use FlatButton	Pass

7. Automated GUI tests (test_room_gui.py)

These run on a machine with a display. **Record the result from your own run here:** [Pass / Fail, date].

ID	Test	Verifies	Expected
TC-G-01	full_room_draws_all_12_charms_in_every_theme	Every one of the 12 charms is drawn on the room canvas in all 4 themes; screenshots are saved to test_output/	Pass
TC-G-02	starter_room_hides_locked_charms	Only equipped charms are drawn in the starter room	Pass

ID	Test	Verifies	Expected
TC-G-03	every_visible_widget_has_readable_text	After building Room, Timer, Analytics, Tasks and Calendar in each theme, no label or button has a contrast below 3 : 1	Pass

8. Manual test cases

Fill in the Actual result and Status columns while running the application.

ID	Steps	Expected result	Actual result	Status
TC-M-01	Run python3 main.py	Window opens with no error; six tabs shown		
TC-M-02	Theme Sakura: open Room, Timer, Analytics, Tasks, Calendar, Settings and the Wardrobe	All button and label text clearly readable; no light text on a light background		
TC-M-03	Same walk-through in Matcha	As above		
TC-M-04	Same walk-through in Lavender	As above		
TC-M-05	Same walk-through in Midnight	As above		
TC-M-06	In Settings, change the theme from Sakura to Midnight, then back, without restarting	Navbar, tabs, title, charm pill, Settings page and every tab change colour together; no mixed palettes		
TC-M-07	Look at the navbar title and the charm-count pill in each theme	Both clearly readable		

ID	Steps	Expected result	Actual result	Status
TC-M-08	Run python3 main.py --dev	DEV MODE badge shown; Developer Tools panel in Settings; cozystudy_data_dev.json is created; cozystudy_data.json is unchanged		
TC-M-09	Settings, then Unlock Everything (or press Ctrl+Shift+U); open Room	Room shows all 12 charms (list in Section 9); Wardrobe shows every charm as Equipped; balance is 9999		
TC-M-10	After unlock, open Analytics	KPI cards, donut chart, 7 bars and 3 timeline blocks for today all have data		
TC-M-11	After unlock, open Calendar and click today	Session dots across the last 14 days; Day Inspector lists 3 sessions		
TC-M-12	After unlock, open Tasks (All Active, then Completed)	Overdue, Due Today and future deadlines shown; High, Medium and Low badges readable; 2 completed tasks		
TC-M-13	Settings, then Reset to Demo	Room returns to the 5 starter charms; theme is kept		
TC-M-14	Click +1000 Charms, Sim 25-min (95%) and Sim Distracted (50%)	Balance rises; 2 new sessions appear in Analytics and Calendar; the 50% session shows as Tab Switching Alert		
TC-M-15	Run python3 main.py (without --dev)	No DEV badge; no Developer Tools panel; Ctrl+Shift+U does nothing		
TC-M-16	Honest mode: start a focus session, switch to another app for about 10 seconds, return	Timer keeps running; tab switches goes up; off-focus seconds shown; integrity below 100 percent		
TC-M-17	Strict mode: start a focus session and switch away	Session pauses; warning dialog; button reads Resume; sound plays if enabled		

ID	Steps	Expected result	Actual result	Status
TC-M-18	Change Anti-Cheat mode in Settings, then open the Timer tab; restart the app	Timer drop-down shows the same mode; the choice is still set after restart		
TC-M-19	Add a task, click Focus Now, then Done	Task is bound to the timer; completing gives a bonus message of 20 to 45 charms; completed tasks have no Done button		
TC-M-20	In Wardrobe, try to unlock a charm you cannot afford, then one you can	Short on charms shows a message; affordable charm unlocks, balance drops by its cost, charm appears in the room		
TC-M-21	Click the cat (Mochi) and the desk lamp in the Room	Speech bubble appears; lamp light toggles		
TC-M-22	Complete data changes, close the app and reopen	Coins, tasks, sessions, charms and theme are all remembered		
TC-M-23	Export Backup (JSON) in Settings	Save dialog opens; file is written		

9. End-state test: the fully decorated Cozy Room

This is the test requested by the reviewer.

Procedure: run `python3 main.py --dev`, open Settings, click **Unlock Everything**, then open the Room tab. Repeat in each of the four themes.

Expected state:

Category	Charms that must be visible
Desk	Steaming Matcha Latte, Pastel Desk Lamp, Baby Succulent Pot, Vintage Lo-Fi Cassette Radio, Amethyst Focus Crystal, Floating Moon Globe
Wall	Twinkling Fairy Lights, Polaroid Memory Board
Pet	Sleeping Calico Cat (Mochi), Fluffy Mochi Bunny

Category	Charms that must be visible
Floor	Giant Monstera Deliciosa, Velvet Cloud Beanbag

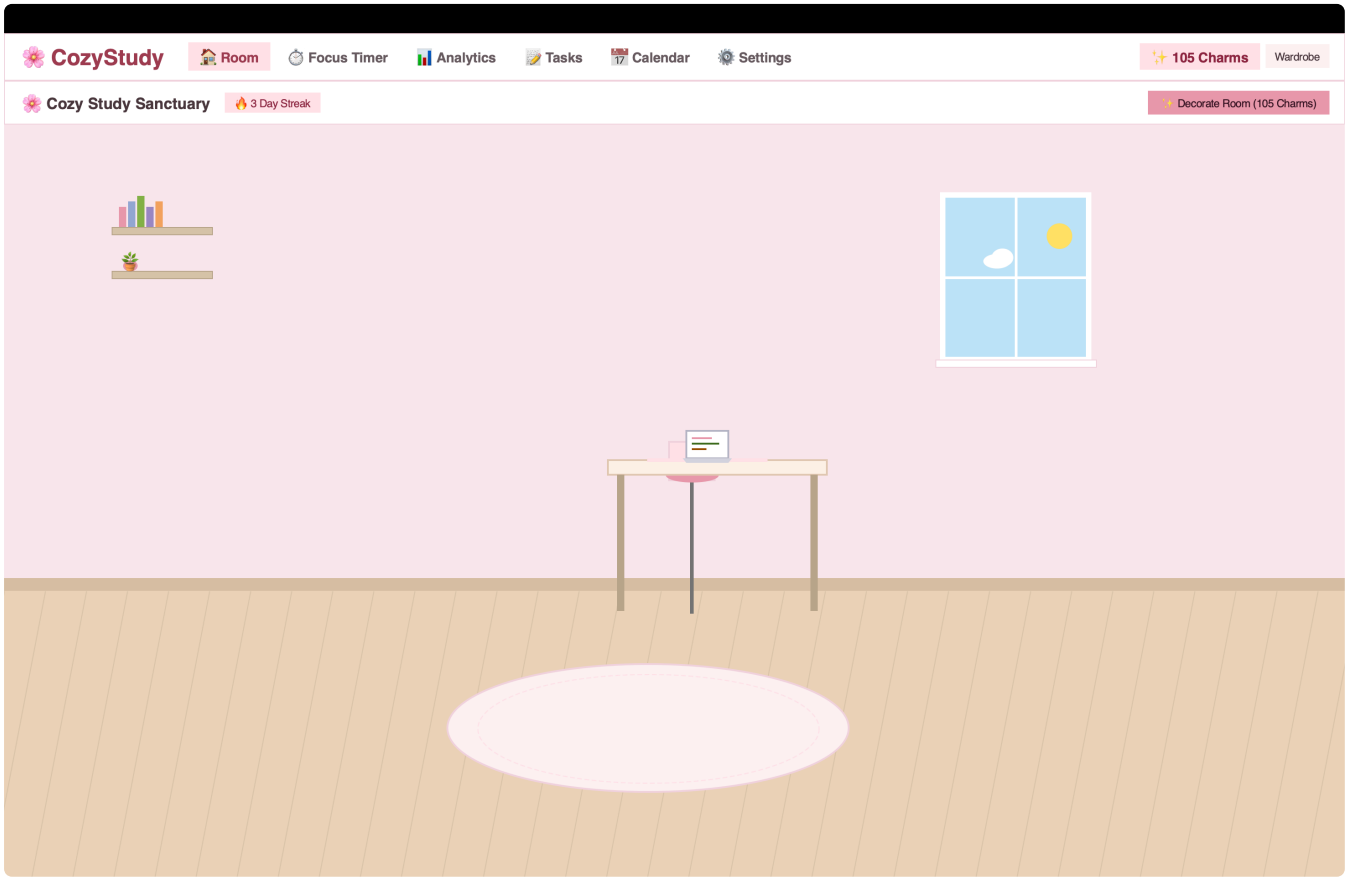
Other expected values: 9999 Star Charms, a 15-day streak (best 21), 31 sessions over 15 calendar days across all 5 subjects, and 7 tasks (2 completed, 1 overdue, 1 due today, the rest in the future).

Backend evidence: TC-B-01 to TC-B-10 and TC-B-23 prove this state without opening a window. **Visual evidence:** TC-G-01 saves a screenshot of the full room for each theme to the test_output folder; attach these to the report.

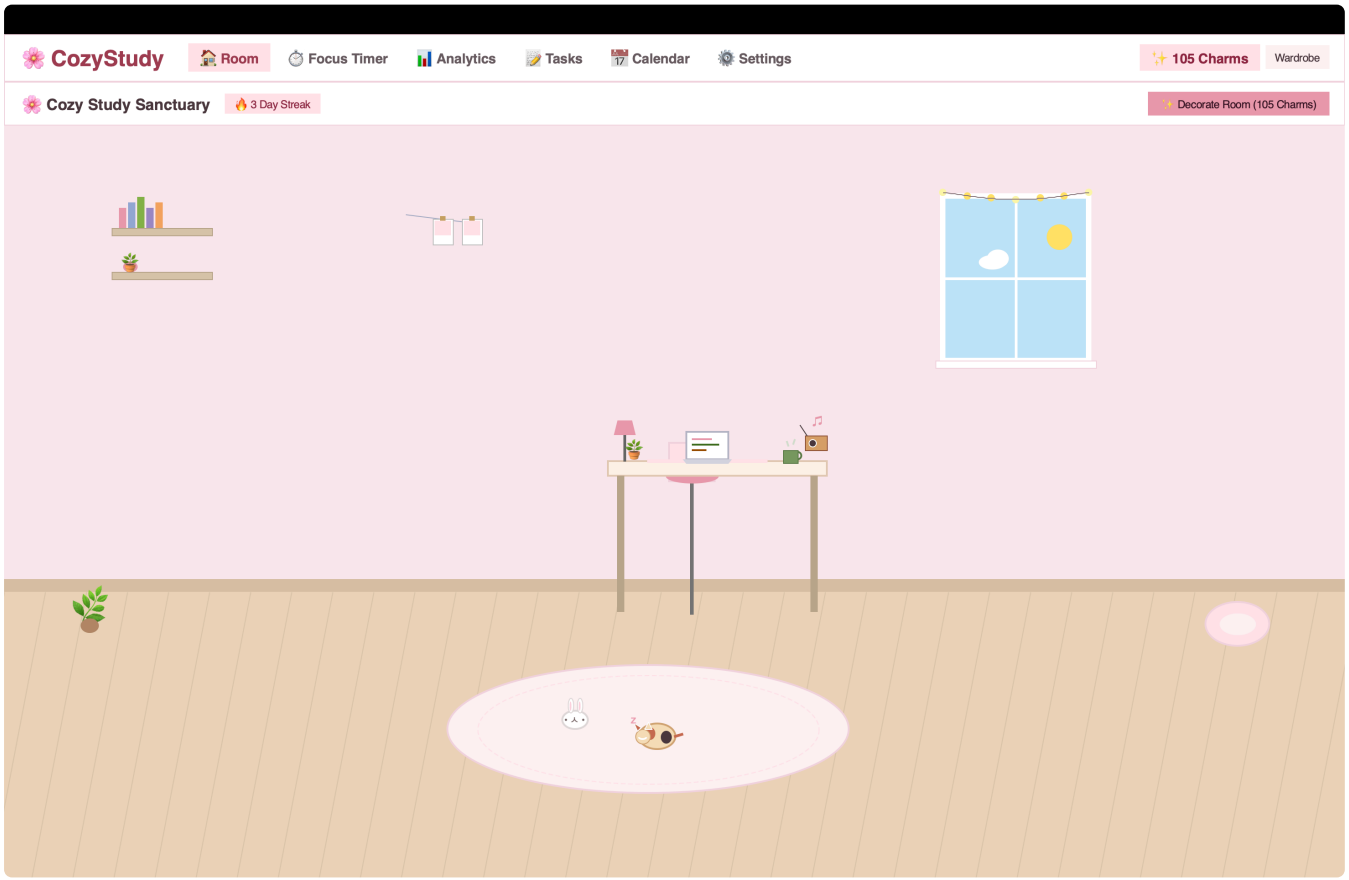
10. Known limitations

- The anti-cheat feature only detects whether the CozyStudy window is in the foreground. It cannot see what the student does in other applications.
- The native drop-down lists and the Wardrobe tab strip use the operating system's colours and are not themed.
- The Done button label always says +20 charms, but the real bonus is 20 to 45.
- The three GUI tests need a display and are not run in a headless environment.

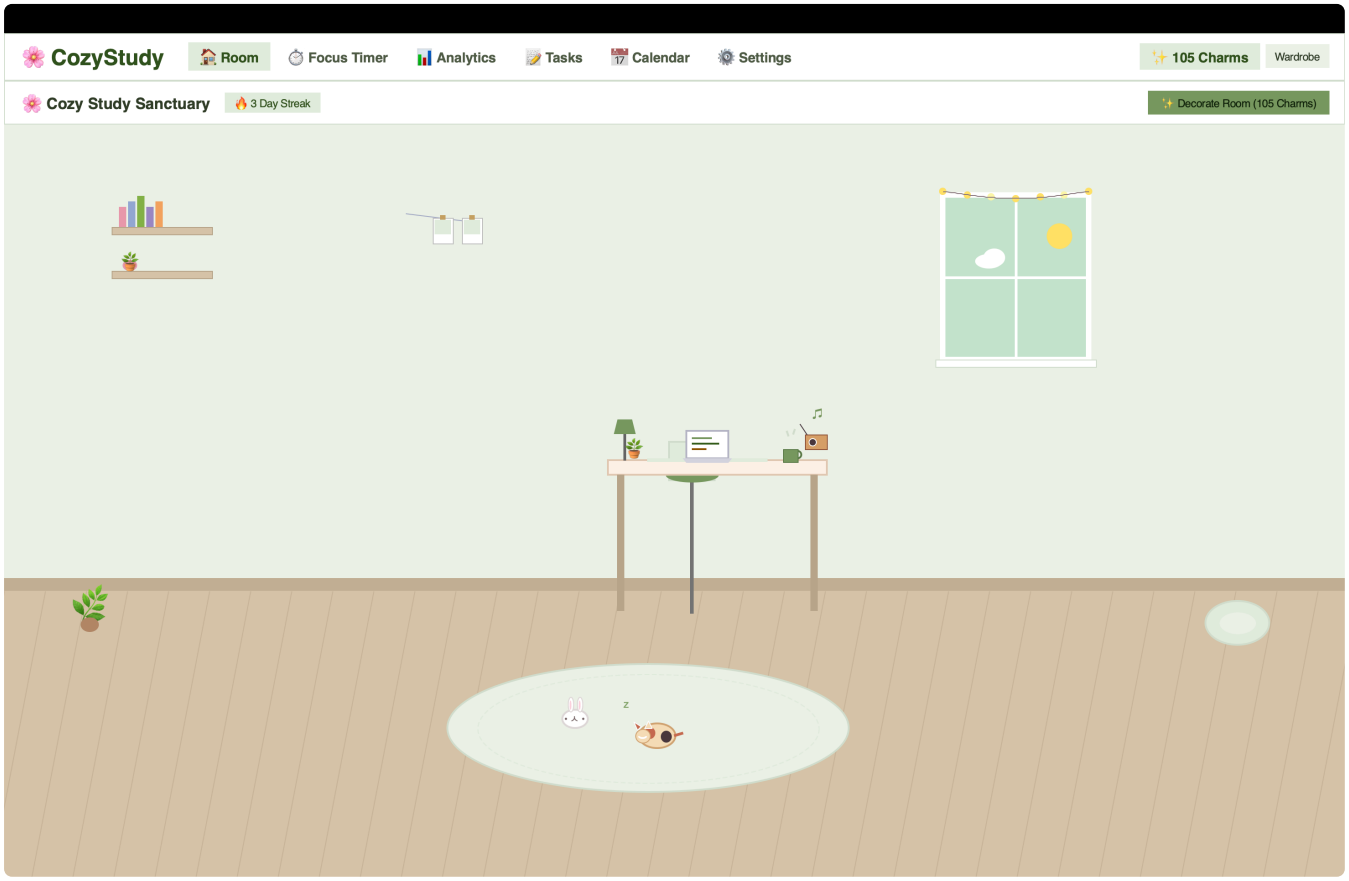
11. Summary



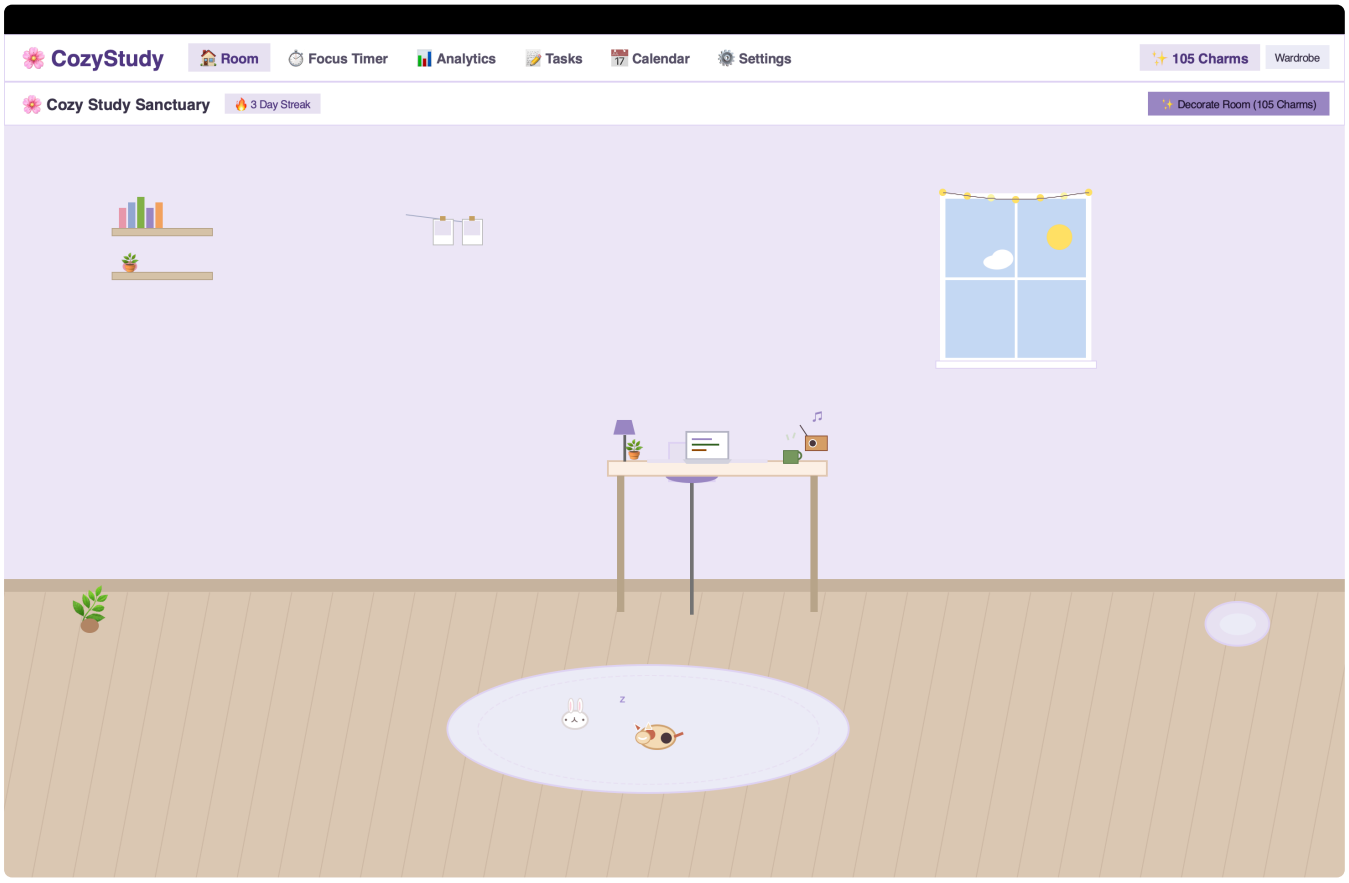
Original Interface (No add-ons)



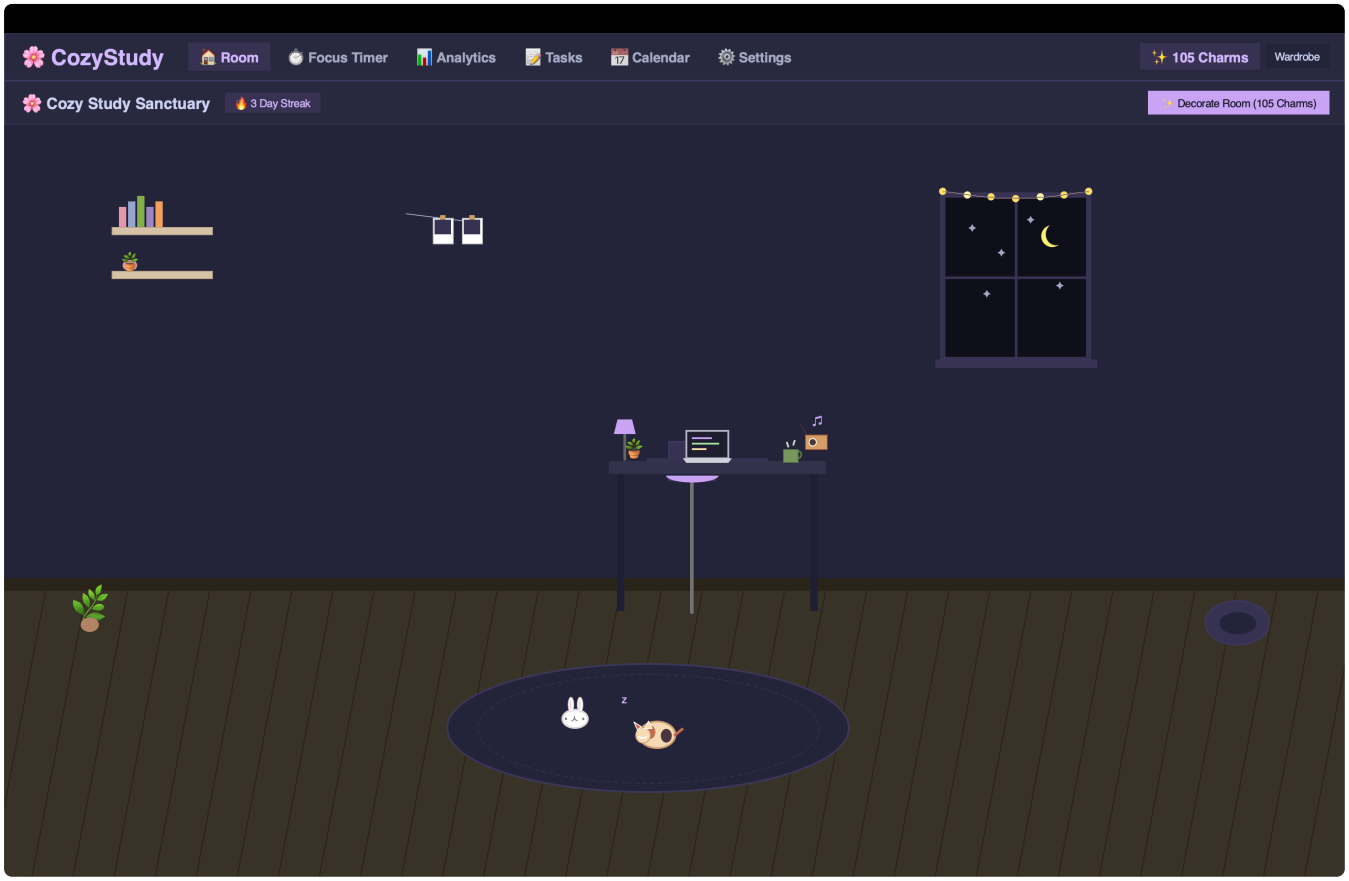
All add-ons equipped (Theme- Sakura)



Theme- Matcha



Theme- Lavender



Theme- Midnight

Item	Result
Automated backend and contrast tests	34 of 34 passed
Colour-contrast audit	0 failures in 524 checks (6 failures in the original code)
Automated GUI tests	[fill in from your run]
Manual test cases	[passed] of 23
Defects found and fixed	7

Prepared by: Vamika Singh A024

Debangi Ray Swain A028

Reviewed by: Mr. Tirup Parmar

Date: 23/10/2026